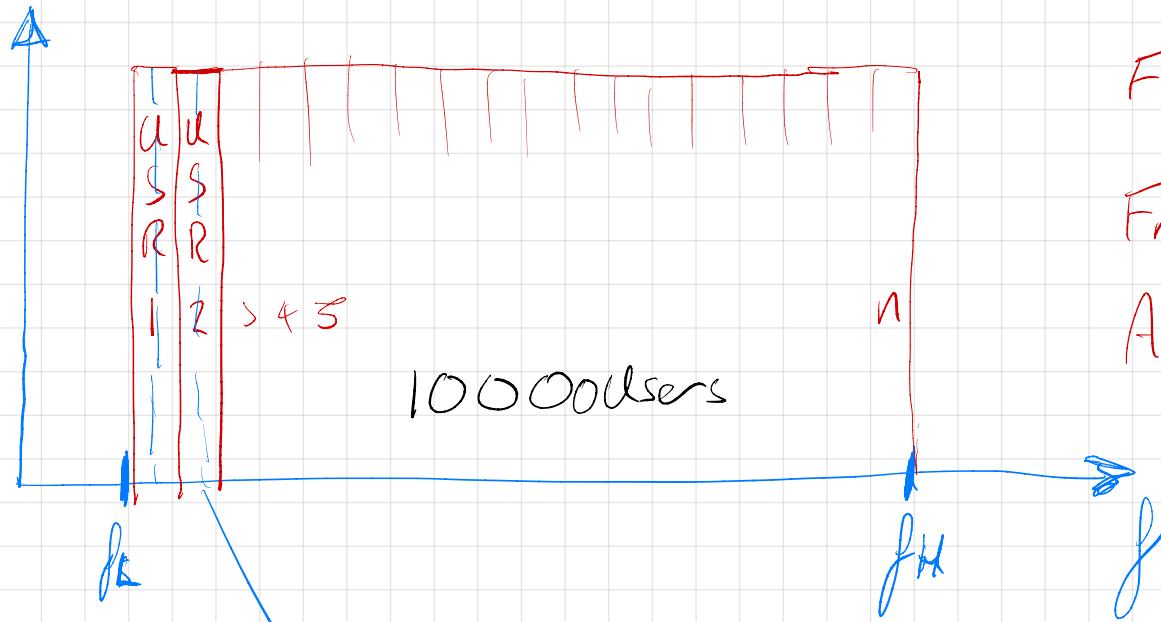


Cellular Systems

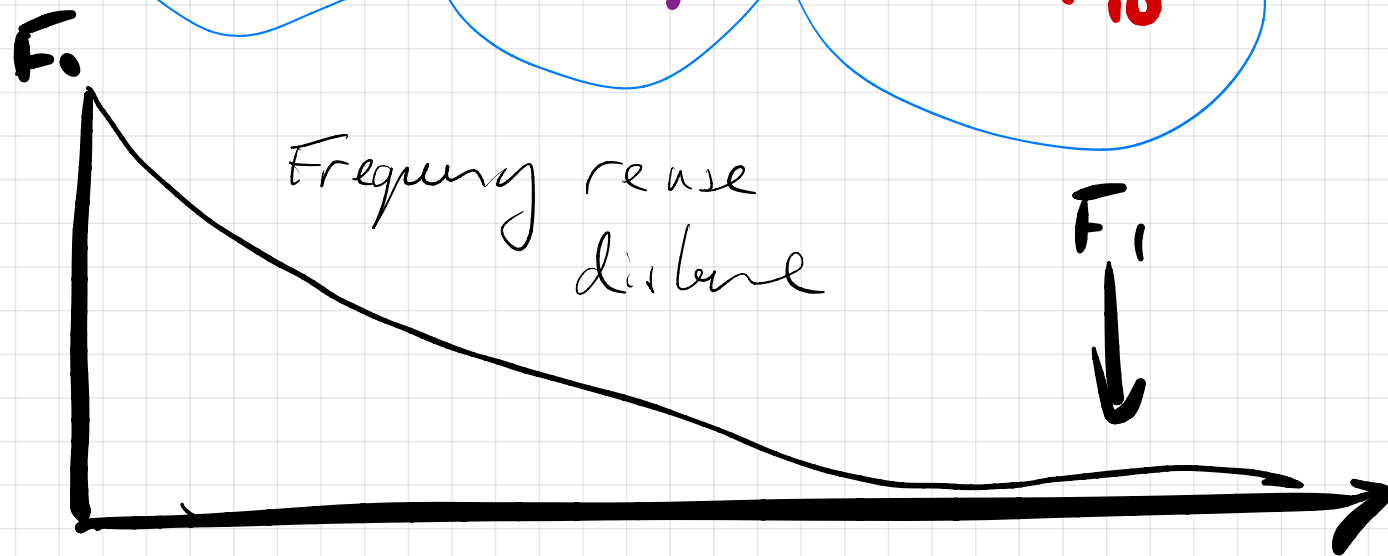
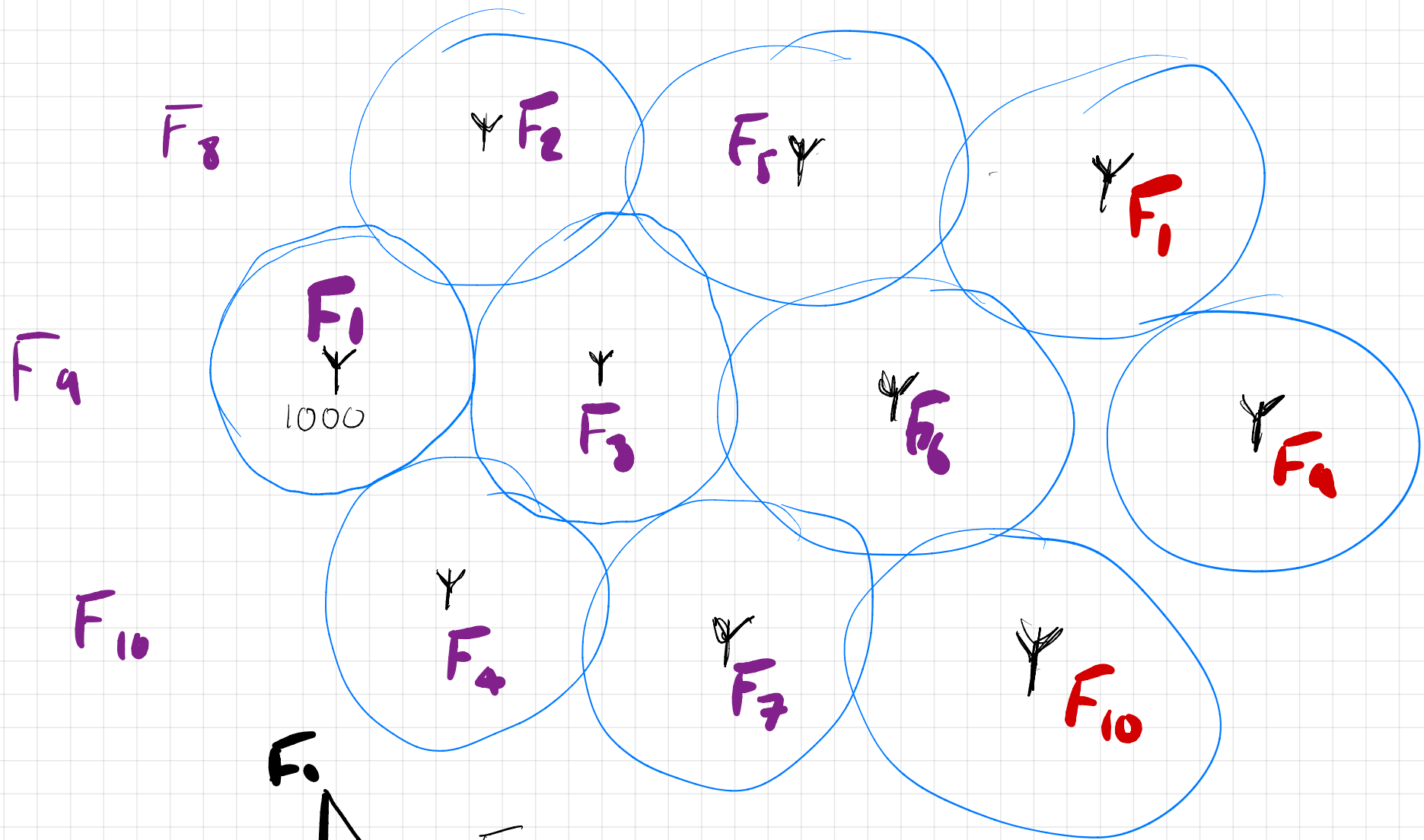
Orange now EE Bought some radio bw £46m



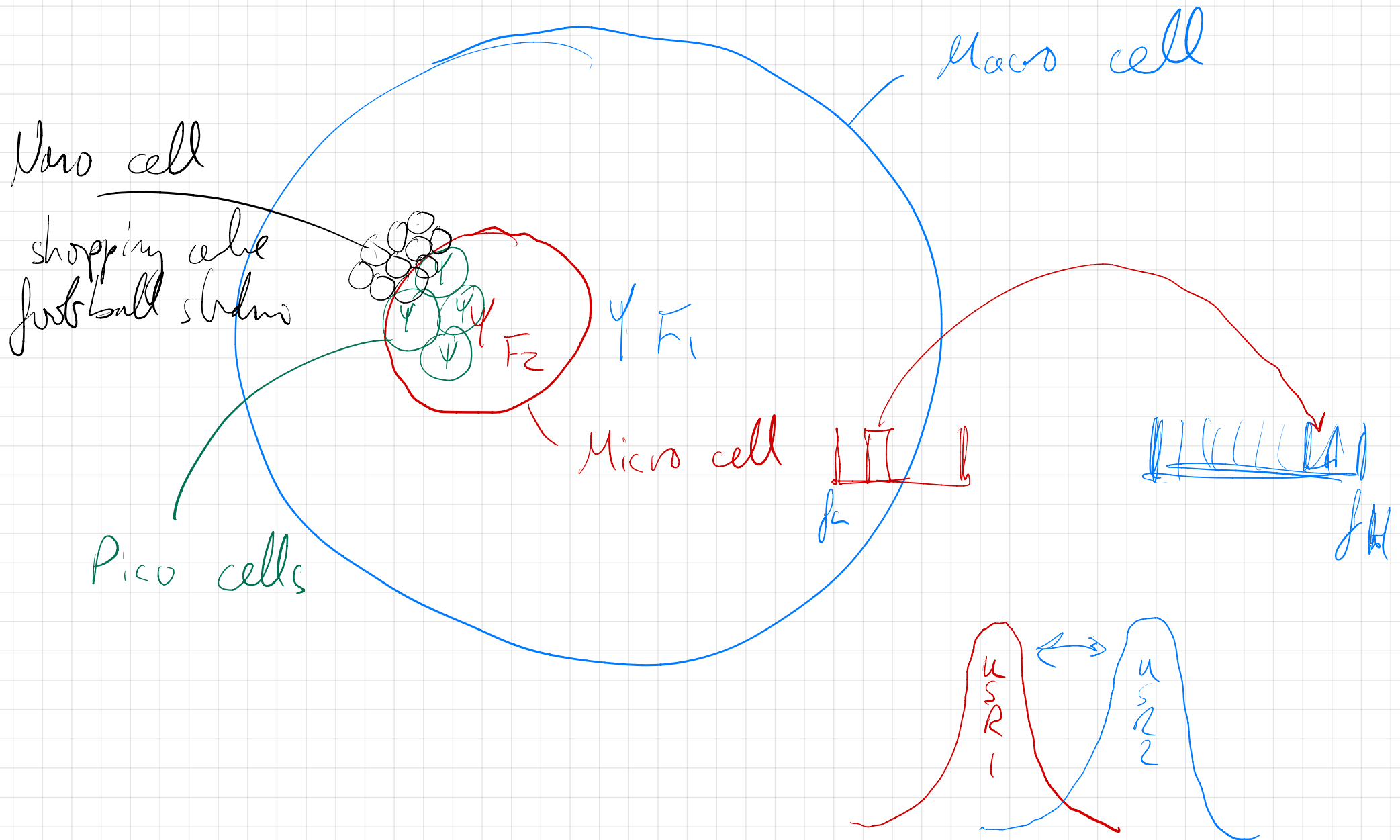
FDMA

Frequency Division Multiple
Access

Each channel is split between uplink & down link
FDD

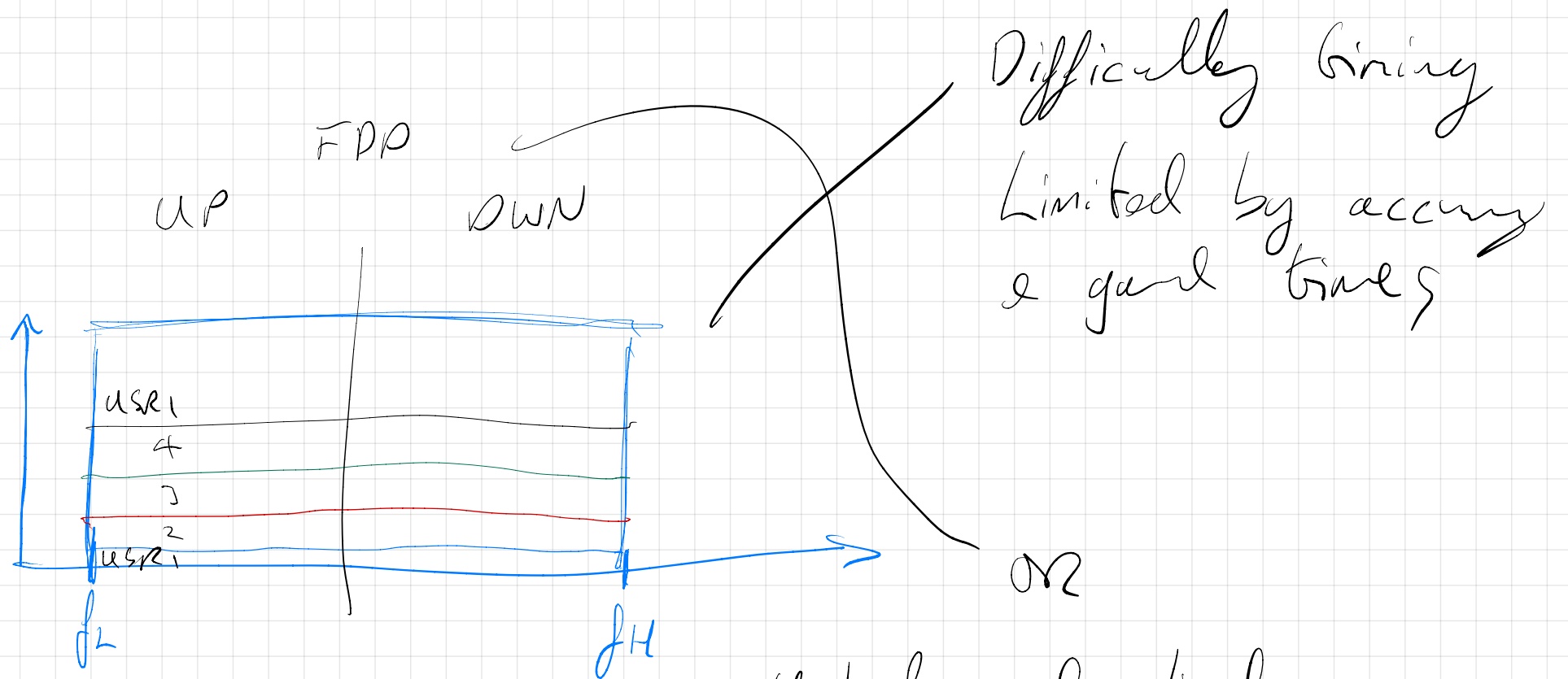


Macro cell $\sim 10\text{km}$ Big cell, rural.



TDMA

Time Division Multiple Access

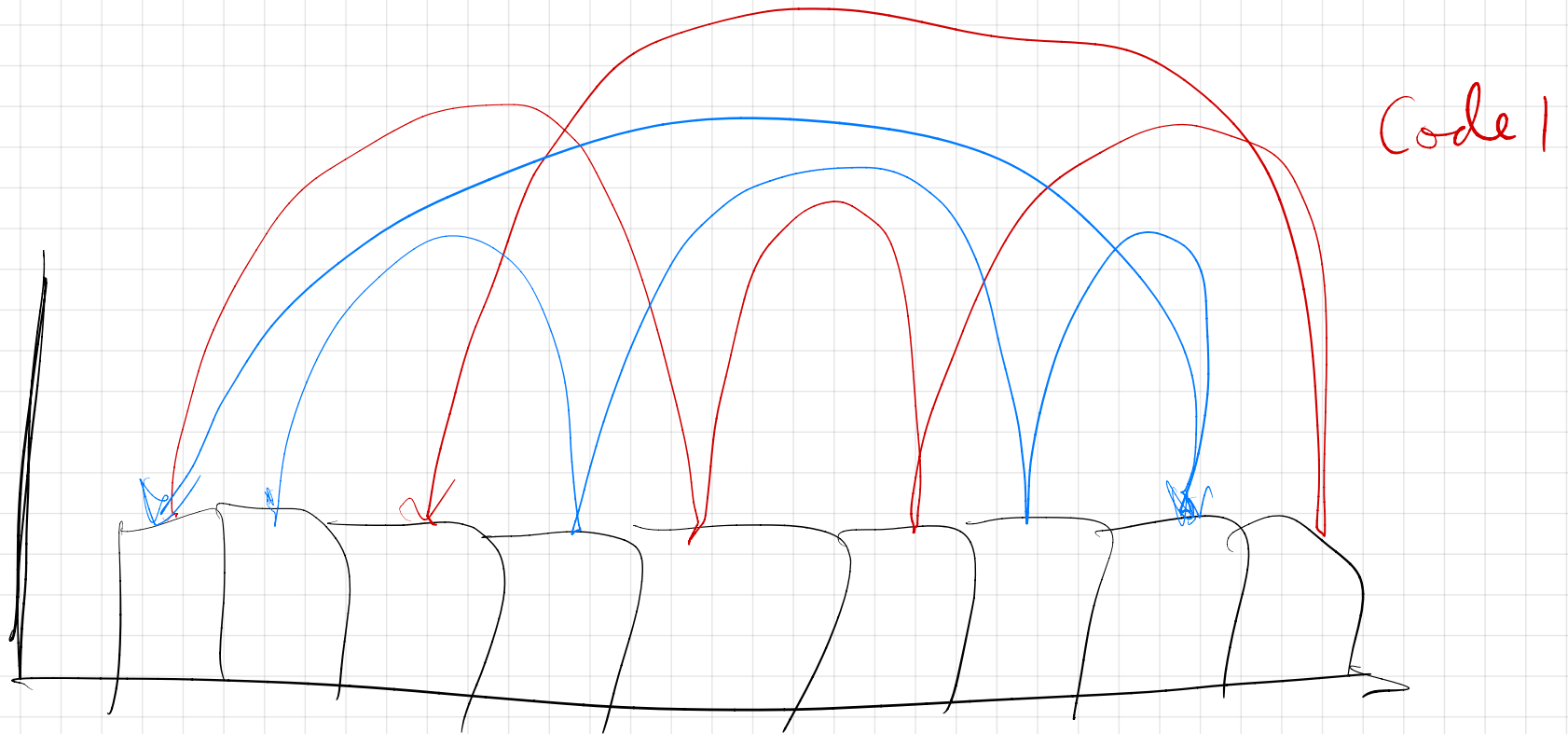


Uplink & Downlink

TDD Time division Duplex

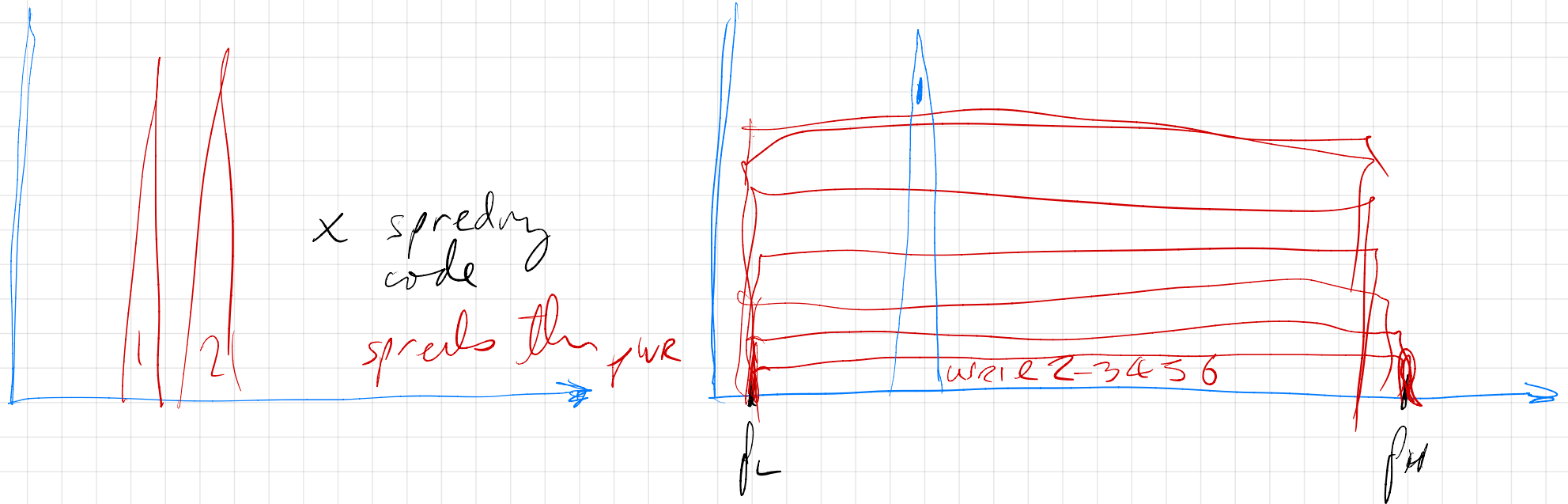
CDMA - FH

Code Division Multiple Access
Frequency Hopping



CDMA - SS

Spread Spectrum
Unique spreading code pseudo random



Cocktail Party Analogy

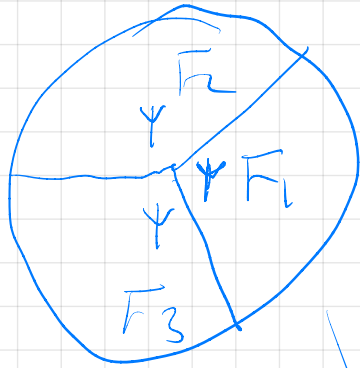
FDMA ~ Big house party, lots of rooms - you have 1 each where you go to talk

TDMA - Bit of a warehouse rare. Take turns to talk & talk for

CDMA-FH - Many room in party house. gone wandering around.

CDMA-SS - Warehouse party - but... instant - lots of diff't languages

r_i



Sectorial Cell.

3x capacity

Better re use distance